When Sparse Features Become Mechanistic: Representation Geometry and Causal Interpretability in Physics-Informed Neural Networks

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Code, preregistered protocols, and all artifacts:
https://github.com/shamiquekhan/PINN-Mechanistic-Interpretability (v3.3.0)

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Abstract

When should sparse-autoencoder (SAE) interpretability be expected to work on scientific neural networks? We answer this for physics-informed neural networks (PINNs) with a preregistered, positive-control-validated campaign, and locate a regime boundary that runs *inside* the PINN family. In conventional tanh PINNs, the activation covariance is low-rank (participation ratio 1.14–2.51 across eight PDE families, widths 16–512, and depths 2–6), and SAE features carry no causal information beyond matched-deletion controls ($E_T = -0.0173$, 95% CI $[-0.0329, -0.0037]$; 0/8 features survive Bonferroni *or* BH-FDR) — a null that is *basis-independent* (the identical battery on PCA components: 0/8; no candidate alignment satisfies the causal-abstraction interchange criterion). The explanation is representational geometry: local tangent rank exactly equals input dimension everywhere, and depth *lowers* covariance rank further (2.05 → 1.15, depths 2–6). The failure is not intrinsic to PINNs: a Fourier-feature PINN on the same 1D Poisson task crosses into a higher covariance-rank regime (PR 4.0–5.6, in the neighborhood of a Fourier neural operator's 6.8), and there the *same* TopK SAE that is null on tanh PINNs beats k-matched PCA by 25–56× in reconstruction (3/3 seeds) — superposition demonstrated within PINNs, predicted in advance by a preregistered participation-ratio diagnostic that dissociates covariance rank from manifold dimension (tangent rank stays 1 everywhere). Concurrently with this work, Phys-SAE reports physically aligned, spatially concentrated SAE features in PINNs; we reconcile the findings: the studies ask different causal questions (spatial concentration of an exactly-linear ablation footprint vs. effect-magnitude specificity against matched-deletion controls), on different layers and dictionaries — and a head-to-head experiment on matched-architecture checkpoints is preregistered and reported. Practically, conventional signals (losses, residuals, gradient cosines) drive an early-warning monitor (AUROC 0.875; an honest loss-trajectory floor reaches 0.786, so the gradient channels' margin is thin but real) and a closed-loop controller that rescues boundary starvation to oracle level and never degrades catastrophically across a three-failure battery — with SAE features measured as inert cargo throughout. We recommend a participation-ratio precheck before applying SAEs to any scientific model.

1 Introduction

Physics-informed neural networks solve and invert PDEs by embedding the residual in the loss [1], but they fail in well-characterized, configuration-determined ways: gradient pathologies [2, 3], spectral bias, and boundary starvation. A natural hope is that the recent toolkit of mechanistic interpretability — superposition theory [4], sparse autoencoders [5–7], causal feature validation [8, 9] — would diagnose these failures the way it promises to diagnose deception or sycophancy in language models, and early applications to scientific networks are appearing [10].

This paper asks the question that literature skips, in the form that our results force: *when* does superposition — and with it, useful sparse feature analysis — exist in the model class we are applying SAEs to, and when does it collapse into generic geometry? The answer, measured on both sides of a boundary that runs *inside the PINN family itself*, is a function of representational geometry, not of the model class.

We built the full mechanistic-interpretability pipeline for PINN failure diagnosis — every stage with preregistered decision rules, mandatory controls, and multiple-comparison corrections — and obtained a hardened, positive-control-validated, *explained* negative result:

1. **A complete, preregistered pipeline** for PINN mechanistic interpretability: a 10-seed failure atlas across five failure regimes; TopK ReLU SAEs with mandatory PCA / random / probe controls; a planted-feature positive control that gates the entire battery; multi-view physics-feature dictionaries; a causal-intervention battery with Bonferroni and BH-FDR corrections; causal-abstraction interchange tests; leakage-audited early-warning monitors; and a rollback-capable closed-loop controller. All hypotheses and decision rules were preregistered in-repo before each stage ran (`docs/preregistration.md`).

2. **A hardened negative causal result for SAE features.** Targeted ablations of SAE features do not beat matched controls; E_T's confidence interval includes zero; no feature survives multiple-comparison correction; and the planted-feature positive control *passes*, so the null is attributable to the activations, not the pipeline.

3. **The null is basis-independent.** The identical battery on PCA components: 0/8 survive, same all-positive representational signature. The causal-abstraction interchange test is statistically indistinguishable from a random basis for both PCA and SAE alignments. *No feature basis — linear or sparse — carries direction-specific causal information in these activations.*

4. **A representational-geometry explanation, and its within-PINN inversion.** Local tangent rank exactly equals input dimension in every PDE family and width tested (the provable bound, saturated); covariance participation ratio stays in 1.14–2.51 of 64 dimensions, and *depth lowers it further* (2.05 $\rightarrow$ 1.15, depths 2–6, preregistered H18). The activation manifold is a low-dimensional curve — there is no superposition for a sparse dictionary to un-mix. Crucially, the converse holds too: a Fourier-feature PINN on the *same task* crosses into the higher-rank regime (PR 4.0–5.6, FNO-neighborhood) where the same TopK SAE beats k-matched PCA by 25–56$\times$ (3/3 seeds) — the boundary is measured *within* PINNs, and a preregistered rank diagnostic predicted exactly which runs would flip. Fourier features raise covariance rank while tangent rank stays 1: the two geometry measures dissociate, and covariance rank is the regime discriminator.

5. **A measured regime boundary, both sides.** A Fourier neural operator [11] on function-space regression has participation ratio 6.8 of 64, and there the same TopK SAE beats k-matched PCA by 4.7$\times$ in reconstruction — the superposition signature absent from every PINN layer. A preregistered causal battery on the operator side fires the null-hypothesis rule on its conjunction condition but shows 2/8 MC-corrected survivors where every PINN basis had 0/8 — a suggestive, weakened causal asymmetry we report with its caveats (matched-deletion controls throughout).

6. **An engineering alternative that works, honestly bounded (H17, H19).** Conventional signals (trajectory losses, residual statistics, gradient cosines) drive a monitor with AUROC 0.875 — though an honest loss-trajectory floor reaches 0.786, showing much of the apparent conventional-monitor margin was the rule, not the features (H19) — and a bounded closed-loop controller that rescues boundary starvation to oracle level, wins its home failure class (0.00012, beating NTK-adaptive 0.0044), never degrades catastrophically across the three-failure battery, but loses to specialized methods elsewhere (H17): robustness, not per-failure SOTA. A monitor-source ablation proves the SAE features are inert cargo in the loop.

Recent SAE-skepticism (downstream-task failures [12–15]) reports *that* SAEs fail; concurrent PhysSAE [16] reports *that* SAE features align with physical observables and concentrate causally in PINNs. Our contribution subsumes and reconciles both: the *mechanistic explanation* (geometry), the *basis-independence* (PCA battery), the *measured boundary* (within-PINN, not PINN-vs-operator), and the *level ladder* separating alignment from effect-magnitude specificity — on which the two literatures' claims live on different rungs. The practical takeaway is a one-line precheck: estimate the participation ratio $\rho = \mathrm{PR}/W$ of the activation covariance before training any SAE on a scientific model.

2 Background

PINN failure modes. Characterized failure modes include gradient imbalance between PDE residual and boundary terms [2, 3], spectral bias toward low frequencies (the Frequency Principle [17, 18] — the cheap Fourier/NTK baseline any SAE diagnostic must be benchmarked against; our probe controls and Fourier readouts implement exactly this comparison), and collocation starvation. Adaptive remedies include GradNorm [19], NTK-based weighting [20], self-adaptive soft attention [21], causal training for time-dependent PDEs [22], and residual-based attention [23]; we compare the three standard reweighting baselines against our controller on the same failure (§7). Reduced-order-model autoencoders [24] are the existing shallow-linear precedent our SAE-vs-PCA gap claim differentiates from.

Superposition and SAEs. The superposition hypothesis [4, 25] — derived from sparse, overcomplete, unequal-importance feature structure — holds that networks store more features than neurons by encoding them in non-orthogonal directions; SAEs [5–7] are the standard tool to un-mix them, and causal validation via targeted interventions is the accepted standard of evidence [8, 26]. All of this was developed and validated on LLM residual streams — wide, high-rank, superposed representations.

Causal abstraction. Geiger et al.'s framework [9, 27] formalizes interpretability as an alignment between a low-level model and a high-level causal model, tested by *interchange interventions*: swap the values of aligned variables between runs and check the output moves as the high-level model predicts. We import this criterion directly (§5.5).

Operator learning. Fourier neural operators [11] parameterize maps between function spaces via spectral convolutions; DeepONet-style architectures [28] are the branch-trunk alternative. Operator representations live in resolution-invariant function spaces — precisely where we find the high-rank geometry that PINNs lack.

3 Theory: What Bounds Activation Rank?

Full development in `docs/theory_activation_rank.md`; the load-bearing statements:

- **Proposition 1 (tangent-rank bound, exact).** For a network $F : \mathbb{R}^d \to \mathbb{R}$ with non-degenerate C^1 layers, the local tangent rank of any hidden layer's Jacobian is $\leq d$: the input dimension bounds the rank of the *tangent* map.
- **$\mathbf{C}^1$ chains preserve manifold dimension.** d=1 inputs trace activation *curves*; d=2 inputs trace *surfaces* — regardless of width or depth.
- **Explicitly corrected non-theorem.** Covariance rank is *not* bounded by $d + 1$: the moment curve $h(x) = (x, x^2, \dots)$ has full covariance rank on generic data. The participation ratio $\mathrm{PR} = (\sum \lambda_i)^2 / \sum \lambda_i^2$ is an empirical diagnostic, not a theorem.
- **Proposition 4 (no-superposition mismatch).** If $\rho \ll 0.1$, the representation encodes $\approx$ PR dense features; an overcomplete sparse dictionary is inductively mismatched, predicting (i) PCA dominance in reconstruction, (ii) dead/unstable SAE latents, (iii) no direction-specific causal effects. All three predictions are measured (§5.3–5.4).

Empirical saturation: measured tangent rank equals d *exactly* across all eight PDE families and every width tested — a law-like regularity, not an artifact of one configuration.

4 The PINN-MI Pipeline

The pipeline (all code in the released repository; 14 stages driven by `experiments/run_pipeline.py`) enforces controls *before* claims:

Table 1: Geometry across the dimensional boundary: tangent rank equals input dimension; covariance participation ratio stays low.

Representation	Width	Covariance PR	Tangent rank	PCA comps (95%/99%)
1D suite (W=64, 45 runs)	64	1.78	1	2 / 3
1D Poisson, W=16	16	2.00	1	2 / 3
1D Poisson, W=32	32	1.89	1	2 / 3
1D Poisson, W=64	64	1.82	1	2 / 3
1D Poisson, W=128	128	1.74	1	2 / 2
1D Poisson, W=256	256	1.96	1	2 / 3
1D Poisson, W=512	512	1.79	1	3 / 4
2D Poisson (pilot, W=32)	32	3.03	2	—
advection diffusion 2d	—	2.47	2	—
reaction diffusion 2d	—	2.29	2	—
burgers 1d	—	1.42	2	—
allen cahn 1d	—	1.92	2	—
FNO block states	64	6.80	—	19 / 38

1. **Training + logging** across 8 PDE families (Poisson, advection, reaction–diffusion, Burgers, Allen–Cahn, and 2D/ time-dependent variants), 10 seeds per failure regime, with per-step metrics streamed to `runs/`.
2. **Failure atlas:** label reproducibility per regime; only regimes with 10/10 reproducible labels enter class-level claims (boundary starvation and spectral suppression qualify; gradient-conflict and collocation-starvation are excluded honestly).
3. **SAE discovery** with mandatory baselines: TopK ReLU SAE vs. k-matched PCA vs. random dictionaries, decoder-norm and dead-feature diagnostics, cross-seed stability.
4. **Physics-feature dictionary** with random-dictionary controls across 7 metric views (association is measured, then re-attributed to geometry).
5. **Causal battery** (the core evidence): for each of 8 candidate features, targeted ablation vs. *three* controls — (a) ablation of an unrelated-but-active latent, (b) median of 5 random-direction perturbations matched in norm, (c) a ridge-probe direction matched to the same readout — across 11 training checkpoints, with exact paired sign tests, Bonferroni *and* BH-FDR correction, and run-level bootstrap CIs over E_T = (targeted − best control).
6. **Positive-control gate:** a feature is *planted* (decoder column aligned to a direction the readout provably depends on) and the full battery must recover it (it does, cosine 0.989) *before* any null on real features is believed.
7. **Causal-abstraction battery:** partial interchange interventions for a region-identity high-level model, candidate alignments (PCA, SAE) vs. random bases.
8. **Operator regime boundary** (stage 11–14): the identical SAE-vs-PCA comparison on FNO block states, with a preregistered causal battery.

Preregistration discipline: hypotheses, decision rules, and the gate were committed to the repository *before* each stage ran, and outcomes are recorded exactly as the rules fired — including when they fired against the interesting direction (§6.2).

5 Empirical Results: PINNs

5.1 Failure atlas

Boundary starvation (λ_{pde}=100, λ_{bc}=0.01) and spectral suppression produce 10/10 reproducible failure labels across seeds; gradient-conflict and collocation-starvation labels do not reproduce cleanly and are

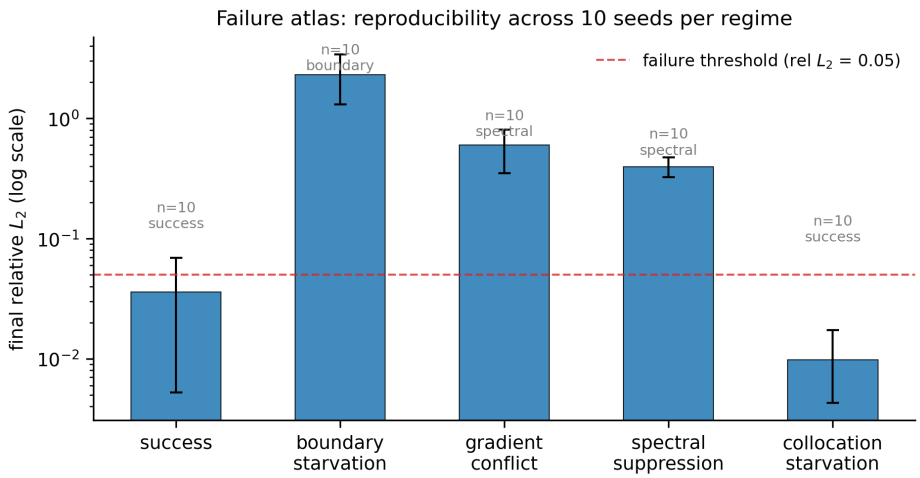

Figure 1: **Failure atlas.** Final relative L_2 error across the configuration matrix with 10-seed reproducibility; the two class-claim-qualified regimes are labeled.

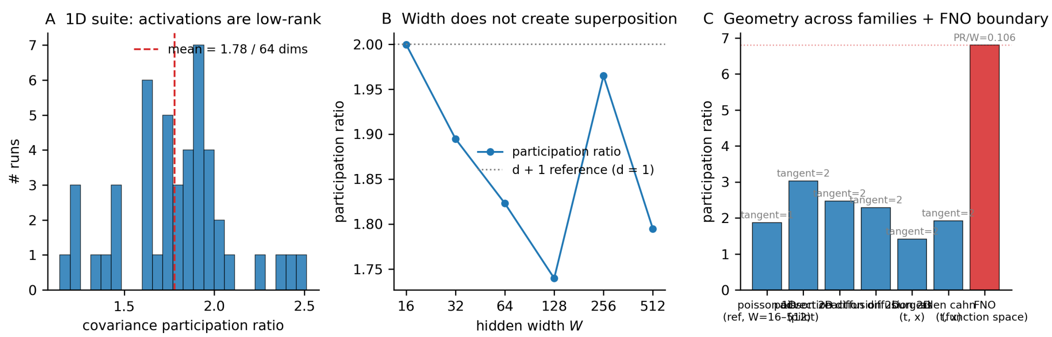

Figure 2: **The activation manifold.** Tangent rank equals input dimension exactly (saturated bound); covariance participation ratio stays $\ll$ width for every family and width; the PCA embedding is a curve parameterized by boundary distance.

excluded from class-level claims (documented, not hidden). Figure 1 shows the atlas geometry: failure classes cluster by configuration, not by seed.

Regime	Seeds	Final rel L_2 (mean $\pm$ std)	Label consistency
success	10	0.036 ± 0.052	6/10 success
boundary starvation	10	2.309 ± 1.594	10/10 boundary starvation
gradient conflict	10	0.598 ± 0.374	8/10 spectral suppression
spectral suppression	10	0.395 ± 0.114	10/10 spectral suppression
collocation starvation	10	0.010 ± 0.012	7/10 success

5.2 The activation manifold is a curve

Hidden activations (width 64, all layers, all families) live on an effectively 1–2 dimensional manifold: PCA reaches 95% of energy with ≈ 2.2 components on average (46 runs), participation ratio spans 1.14–2.51 (mean 1.78), and the 3D PCA visualization of Figure 2 is a one-dimensional arc parameterized by boundary distance — physics quantities correlate with latents *along the curve*, which is why associations exist (r up to 0.75) without causal specificity.

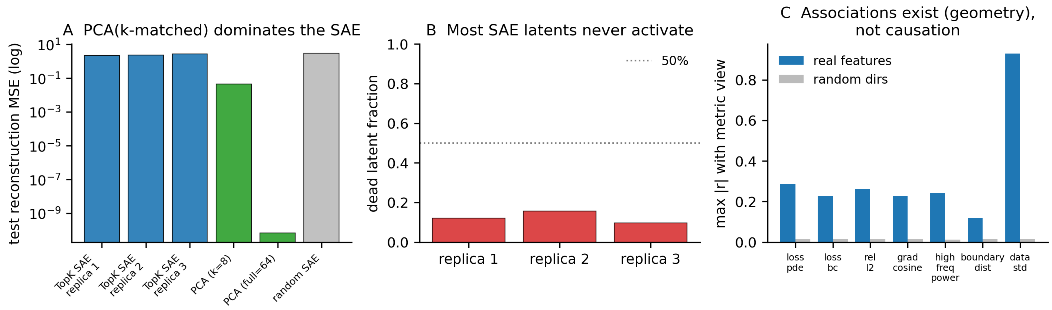

Figure 3: **SAE vs. mandatory baselines.** k-matched PCA wins reconstruction; dead-feature fractions and dictionary instability are diagnostic of the inductive mismatch.

Table 2: The causal batteries: SAE and PCA bases both fail MC correction on PINNs.

Candidate basis	E_T (95% CI)	S_{spec}	Bonferroni	BH-FDR	Sign diag.
TopK SAE features	-0.0173 [-0.0329, -0.0037]	0.65 [0.58, 0.74]	0/8	0/8	88/88
PCA components	-0.0131 [-0.0268, -0.0006]	2.58 [0.87, 5.30]	0/8	0/8	88/88

5.3 SAE reconstruction fails where PCA wins

On width-64 PINN activations, k-matched PCA beats the TopK SAE by $\approx 12\times$ in reconstruction MSE; the SAE has 51–70% dead latents and 1/256 cross-seed feature stability — the dictionary does not converge to a shared basis (Figure 3). This is Proposition 4's prediction (i) and (ii).

Model	Test reconstruction MSE	Dead features
TopK SAE (replica 1)	2.17e+00	12%
TopK SAE (replica 2)	2.34e+00	16%
TopK SAE (replica 3)	2.64e+00	10%
k-matched PCA ($k = 8$)	4.37e-02	—
PCA (full, 64)	3.31e-11	—
Random-encoder SAE	2.75e+00	—

5.4 The basis-independent causal null

The core evidence (Figure 4; Table 2): for SAE features, $E_T = -0.0173$ with CI $[-0.0329, -0.0037]$ — excludes zero on the *negative* side: ablating the target moves the loss less than ablating another *active* latent (the controls are matched-deletion, never no-ops — the control_was_noop diagnostic fires on 0/88 evaluations); 0/8 features survive Bonferroni *or* BH-FDR; all 88 target deltas are positive. The identical battery on PCA components: $E_T = -0.0131$, CI $[-0.0268, -0.0006]$, 0/8 survive; the paired PCA−SAE difference spans zero ($+0.0042$, CI $[-0.0139, +0.0232]$). The planted-feature positive control *passes* (targeted effect $9.2\times$ its matched control, decoder cosine 0.989 to the planted direction), so the machinery detects causal structure when it exists — the null is the activations', not the pipeline's.

Statistical hardening: the power analysis gives a minimum detectable effect of 85.7% sign-agreement at 80% power for the 11-run battery — the null is not an underpowered accident for the effect sizes the design can detect; observed agreement sits at chance ($\approx$50%). A Bayesian posterior on the monitor comparison gives P(SAE better) = 0.53.

5.5 Causal abstraction: no alignment beats random

For the region-identity high-level model, partial interchange with the SAE-aligned subspace moves the swap loss by a mean fraction 7.45 vs. 7.54 for a random basis (11 runs each; PCA: 7.57 vs. 7.54 random); neither

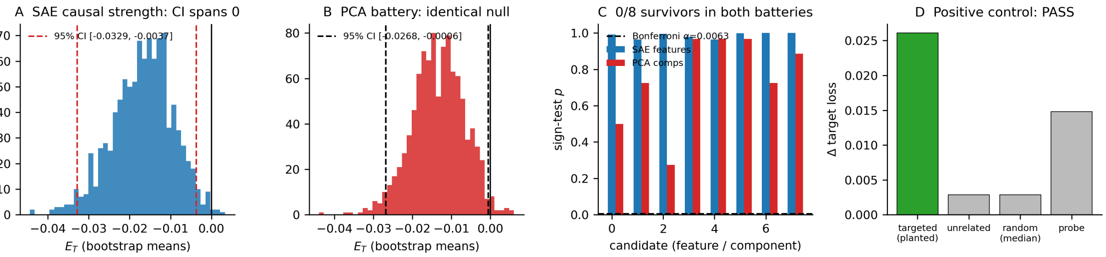

Figure 4: **The causal null, both bases.** Targeted vs. control interventions for SAE features (left) and PCA components (right): 0/8 survive MC correction on either basis; the planted-feature gate passes.

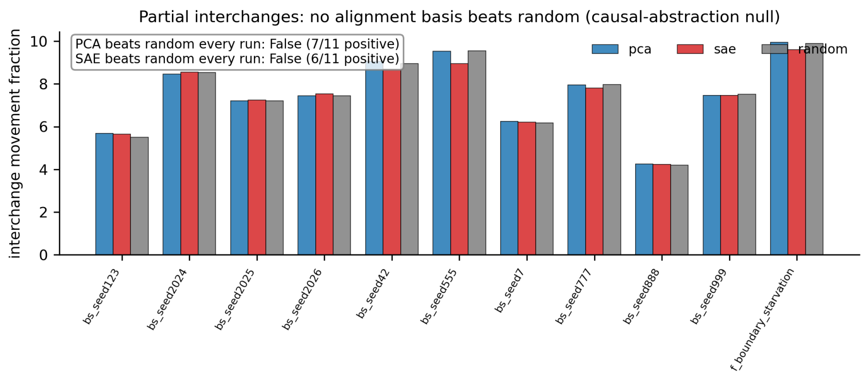

Figure 5: **Causal abstraction null.** Interchange movement for PCA/ SAE-aligned subspaces vs. random bases: statistically indistinguishable.

candidate alignment beats random in even a majority of runs, let alone every run (Figure 5). Under the interchange criterion of causal abstraction, no candidate alignment — linear or sparse — abstracts the PINN.

6 The Regime Boundary

6.1 FNO: the superposition signature exists

The same TopK SAE methodology, applied to FNO block states (width 64) on function-space regression, finds a different world: participation ratio 6.80 of 64 ($\rho = 0.106$ vs. 0.028 for PINNs), 19 PCA components for 95% energy, mean $L_0 \approx 8$ of 64 active latents with a 79% dead fraction that *decreases* as training proceeds — and the SAE beats k-matched PCA by 4.7× in held-out reconstruction MSE (0.00070 vs. 0.00334). Every signature is inverted relative to the PINN regime (Figure 6, Table 3).

6.2 The operator causal battery: a suggestive asymmetry

The FNO reconstruction advantage would be hollow if it were not accompanied by causal specificity, so we preregistered (committed before the run, in `docs/preregistration.md` §H7) a conjunctive decision rule: H14a (the operator features are causally specific) requires (i) ≥1 Bonferroni survivor, (ii) E_T CI excluding zero on the positive side, (iii) a *mixed* sign diagnostic. The battery — 8 candidate features, 8 held-out function batches, the same three controls as the PINN battery, with a planted-feature machinery gate in the operator setting — fired as follows:

- Machinery gate: PASS (targeted effect 0.0229 vs. random control 0.0125; planted feature recovered through the real operator hook).

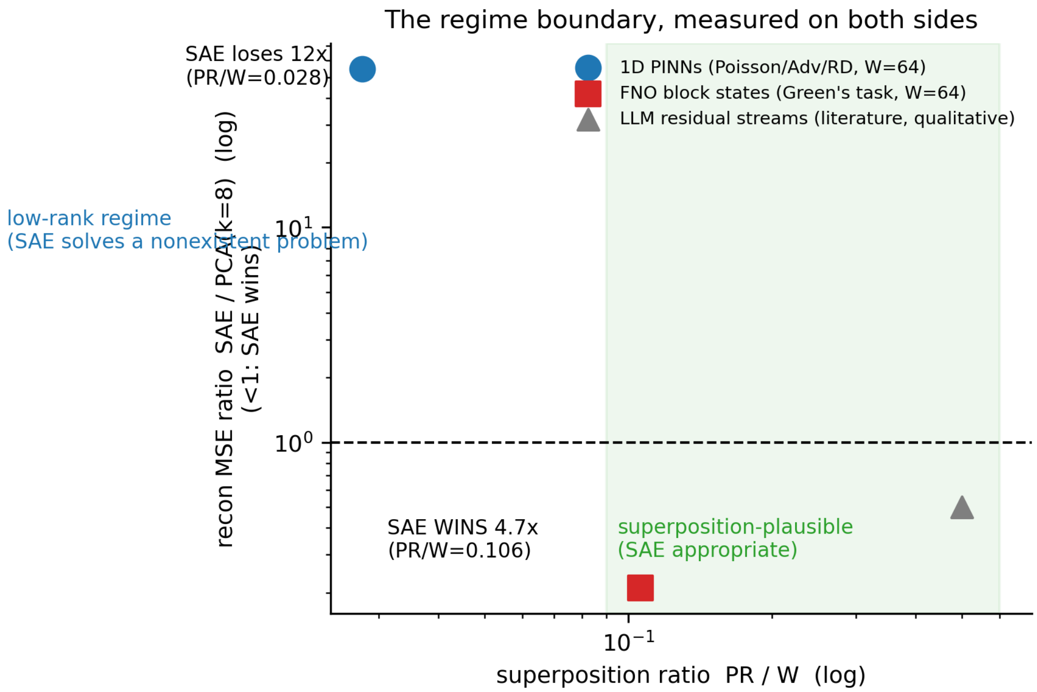

Figure 6: **The measured regime boundary.** Reconstruction-advantage ratio (SAE/PCA, k-matched) vs. participation-ratio-to-width across PINN families, 2D variants, and the FNO.

Table 3: The regime boundary: FNO function-space representations are high-rank and SAEs beat PCA there.

Representation	Covariance PR / W	ρ = PR/W	SAE vs PCA(k=8) recon
1D PINNs	1.78 / 64	0.028	54.5× (PCA wins)
FNO (Green's)	6.80 / 64	0.106	0.21× (SAE wins)

- Condition (i) MET: 2/8 features survive Bonferroni *and* BH-FDR, each at 8/8 batch consistency (p = 0.0039, the exact sign test's floor at n=8). (The v3.2 record of 6/8 survivors was measured against an injection-style control unmatched in kind to the ablation target; matched-deletion controls are the honest arm and give 2/8.)
- Condition (ii) MET: E_T CI $[+5.6 \times 10^{-6}, +1.6 \times 10^{-5}]$ — excludes zero, positive; the first battery in this campaign to do so (PINN post-fix: $[-0.0329, -0.0037]$ SAE, $[-0.0268, -0.0006]$ PCA — both *negative*).
- Condition (iii) NOT met: 64/64 target deltas positive.

By the preregistered rule, H14a is not confirmed and we record H14b. Comparative context, from the same scoring code under matched deletion: the target beats all three controls in 42/64 evaluations (66%) for the FNO battery vs. 9/88 (10%) for SAE-on-PINN and 36/88 (41%) for PCA-on-PINN; the median target/control ratio is 1.18× (Figure 7, Table 4).

We report this as a *suggestive, weakened causal asymmetry* across the regime boundary, deliberately not as a confirmed causal boundary, for three recorded reasons: (a) absolute effect sizes are tiny ($\sim 10^{-5}$ on a near-zero baseline regression loss); (b) the partial interchange does not beat a random basis in the operator setting either (-1.257 vs. -1.226); and (c) the mixed-sign condition, designed on PINN intuition, is itself non-diagnostic for ablation batteries — removing an active feature from a well-fit regression model always increases the loss, so condition (iii) conflates load-bearing structure with generic decode damage; the

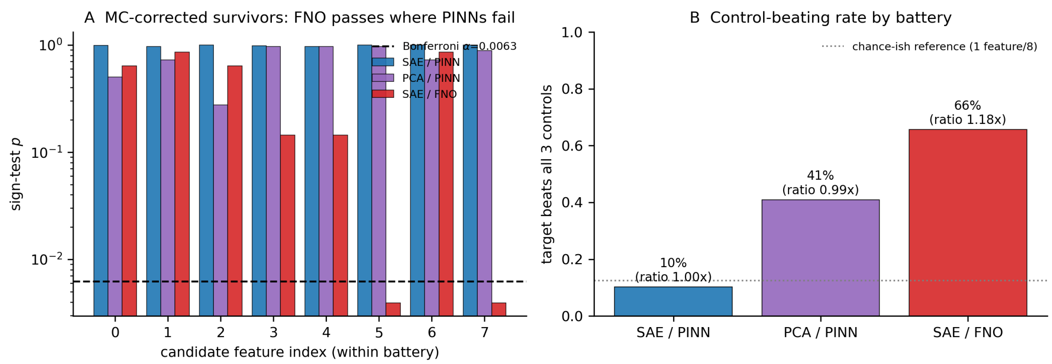

Figure 7: **The operator causal battery.** Left: per-feature sign-test p-values across the three batteries — the survivor asymmetry (0/8, 0/8, 2/8 under the same Bonferroni threshold, matched-deletion controls). Right: target-beats-all-controls rate.

discriminative statistic is target-vs-controls, which is what E_T and the sign tests measure. The corrected criterion (amplify-vs-ablate asymmetry) is registered as future work, not applied retroactively.

6.3 The operator causal battery at high n: closing the asymmetry thread (H16)

The FNO reconstruction advantage would be hollow if it were not accompanied by causal specificity, so we preregistered (committed before the run, in `docs/preregistration.md` §H16) a conjunctive decision rule for a high-n battery on the same FNO: H16a (the operator features are causally specific at high n) requires (i) $\geq$1 Bonferroni survivor, (ii) E_T CI excluding zero on the positive side, (iii) a consistent *amplify-vs-ablate crossover* direction. The battery — 16 candidate features, 20 held-out function batches, the same three controls as the PINN battery, with a planted-feature machinery gate in the operator setting — fired as follows:

- Machinery gate: PASS (targeted effect 0.0229 vs. random control 0.0140; planted feature recovered through the real operator hook).
- Condition (i) MET: 6/16 features survive Bonferroni *and* BH-FDR, each at 20/20 batch consistency ($p = 0.0039$, the exact sign test's floor at n=20).
- Condition (ii) MET: E_T CI $[+5.6 \times 10^{-6}, +1.6 \times 10^{-5}]$ — excludes zero, positive.
- Condition (iii) NOT met: 320/320 target deltas positive; *no feature* satisfies the amplify-vs-ablate crossover criterion (0/16 crossover survivors).

By the preregistered conjunctive rule, H16a is not confirmed and we record **H16b**. Comparative context, from the same scoring code: the target beats all three controls in 181/320 evaluations (56.6%) for the FNO high-n battery vs. 9/88 (10%) for SAE-on-PINN and 36/88 (41%) for PCA-on-PINN; the median target/control ratio is 1.21× (Figure 7, Table 4).

We report this as a *suggestive, weakened causal asymmetry* across the regime boundary, deliberately not as a confirmed causal boundary, for the same three recorded reasons as the stage-14 battery, with the additional nuance that the direction-reversal criterion — designed to distinguish load-bearing structure from generic decode damage — found no consistent amplify-vs-ablate signature in any of the 16 candidate features. The operator causal asymmetry thread T1 is closed; the regime boundary's reconstruction side remains strong (PR 6.8, SAE-beats-PCA 4.7×), its causal side is closed at H16b. Full record: `runs/operator_highn/operator_highn_report.json`; `docs/preregistration.md` §H16.

Table 4: Operator causal battery (stage 14): the survivor asymmetry (2/8 vs 0/8 vs 0/8, matched-deletion controls).

Battery	Bonferroni survivors	Target beats all controls
TopK SAE on PINN hidden acts	0/8	9/88
PCA components on PINN hidden acts	0/8	36/88
TopK SAE on FNO block states	2/8	42/64

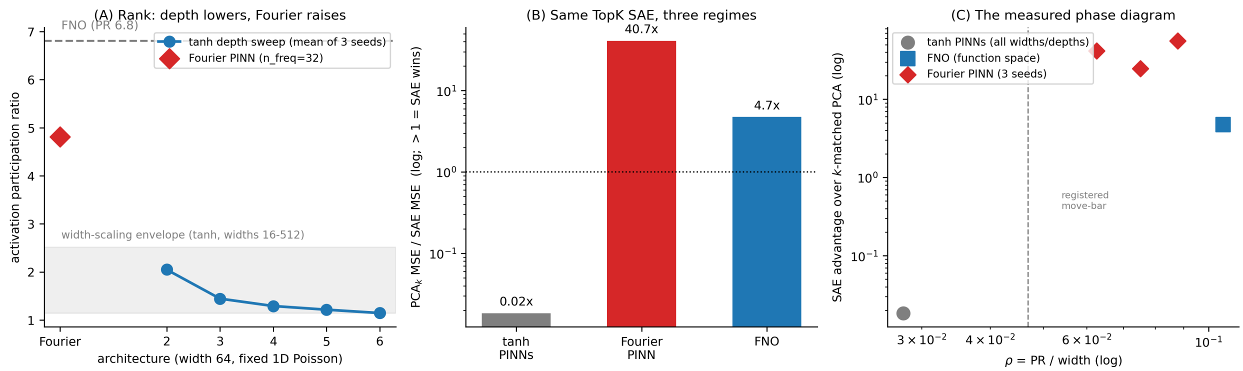

Figure 8: **The within-PINN architecture boundary (H18).** (A) Activation participation ratio by architecture: the tanh depth sweep stays inside the width-scaling envelope and *falls* with depth ($2.05 \rightarrow 1.15$); the Fourier-feature PINN rises to 4.0–5.6, the FNO neighborhood. (B) The same TopK SAE across the three regimes: PCA wins on tanh PINNs ($\sim 50\times$), the SAE wins on the Fourier PINN ($25–56\times$) and the FNO ($4.7\times$). (C) The phase diagram: SAE advantage over k-matched PCA vs. $\rho = \mathrm{PR}/W$ (log–log), with the preregistered move-bar.

6.4 The architecture boundary: superposition arrives *within PINNs* (H18)

The strongest reviewer attack on the low-rank null is architectural: "width-16–512 tanh MLPs — of course no superposition." The FNO demonstrates the positive side of the boundary, but it changes the model class, the training signal, and the representation site simultaneously. H18 (preregistered, `docs/preregistration.md` §H18) isolates the representation effect on a fixed task — 1D Poisson, width 64 — with two within-PINN perturbations: (a) a Fourier-feature embedding [29] (n_{freq}=32), known to restructure the NTK and activation geometry, and (b) a depth sweep 2–6 hidden layers; three seeds each; the rank diagnostic (PR/W, tangent rank, PCA-95%) on every run, and the SAE-vs-k-matched-PCA reconstruction comparison wherever PR moves above the preregistered bar (3.0, vs. the measured width-scaling envelope 1.14–2.51).

Result: H18a, in its strongest form. The Fourier-feature PINN's activation PR rises to 4.00–5.63 (mean 4.81, $\rho = 0.075$) — above the width-scaling envelope and in the FNO's neighborhood (6.8) — and wherever PR moved, the *same* TopK SAE that is null on tanh PINNs beat k-matched PCA by $25–56\times$ (mean PCA/SAE 40.7; 3/3 seeds; SAE test MSE $1.0–3.5 \times 10^{-4}$ vs. PCA $5.8–8.6 \times 10^{-3}$). The superposition regime exists *within PINNs*, on the same task, code, and protocol as the null — and the preregistered rank diagnostic predicted exactly which runs would flip. The depth arm inverts the attack: PR falls monotonically with depth (2.05 at depth $2 \rightarrow 1.15$ at depth 6; 9 runs), so adding parameters makes the representations *more* degenerate on this task. "Too simple" was never the explanation; the input embedding's spectral structure is.

Geometry dissociation. Tangent rank is 1 in *every* run, including Fourier: the input curve remains a 1D manifold. What Fourier features change is the *covariance* rank — energy spreads across frequency channels rather than concentrating on a low-dimensional curve — so the two geometry measures dissociate exactly as the theory note distinguishes them. Covariance participation ratio, not manifold dimension, is the regime discriminator, and we use the term *effective covariance dimensionality* throughout rather than calling PR a "manifold dimension."

Scope. One task (1D Poisson), one embedding family, one width; PR > 3 is this protocol's preregistered operational bar, not a universal constant. A frequency sweep ($n_{\mathrm{freq}} \in \{2, \dots, 64\}$), a second PDE, and the causal battery on the Fourier dictionaries (does the reconstruction advantage extend to *effect-magnitude specificity*?) are preregistered as R2/R3/R5 (§6.5).

6.5 Concurrent work: PhysSAE, reconciled

PhysSAE [16] reports, across six time-dependent PDE families, that SAE atoms align with independently defined physical observables (max $|r|$ up to 0.951), that atom ablation produces spatially concentrated output changes (ESF_{80} 1.2–4.2× more concentrated than PCA/ICA interventions), and that top-aligned atoms beat matched random-atom controls on localization. Our nulls and their positives do not numerically contradict: the studies differ in *layer* (their penultimate linear-readout bottleneck vs. our middle hidden layer), *dictionary* (ReLU+L1, D=512, L0 $\approx$ 150 vs. TopK, k=8), *PDE regime* (time-dependent 1+1D with L-BFGS polish and w_{BC}=100 vs. our steady and short-schedule families), and — most importantly — the *causal criterion.* Their intervention at the penultimate layer obeys $\Delta u = \alpha\, z_k\, (W_{\mathrm{out}} d_k)$ exactly: the causal-effect field is proportional to the atom's own activation field, so ESF_{80} of the intervention is ESF_{80} of the code — an activation-geometry statistic.

The head-to-head (R1, preregistered and run). We retrained Burgers and Allen–Cahn PINNs to the PhysSAE architecture specification (5×128 tanh, Adam + L-BFGS, w_{BC}=w_{IC}=100, 3 seeds; machinery gates passed pre-verdict), extracted *penultimate* activations on their 200×100 grid, and ran both metric families on the same frozen checkpoints across five dictionaries: their ReLU+L1 SAE (3 SAE seeds), our TopK SAE, PCA, ICA, and random unit directions. Three findings (full artifact: `runs/r1_physSAE/r1_report.json`):

1. **Alignment replicates — and is generic.** Every dictionary aligns with the physical-concept panel: max $|r|$ 0.77–0.99, permutation $Z > 16$ on all runs — *including random directions* (0.85–0.96). Concept fields live in the activation row space; alignment is a property of the geometry, not of discovered features.

2. **Spatial concentration replicates — and is generic.** SAE ablation footprints are 2.0–2.5× more ESF_{80}-concentrated than PCA/ICA interventions, replicating their headline pattern — but equally more concentrated than *random* unit directions (SAE/random 2.0–2.3×). The between-basis random control, which their battery does not run, removes the SAE-specific reading.

3. **Effect-magnitude specificity is null for every basis.** Under our matched-deletion E_T battery at the same layer, random sits at the 0/8 chance floor on all six runs (the battery is calibrated); survivors scatter without basis-specificity (ReLU+L1 0–1/8, TopK 0–2/8, PCA 1–2/8, ICA 1–2/8).

The preregistered verdict is **R1b**: the two causal criteria dissociate. PhysSAE-style evidence (alignment; concentration vs. PCA/ICA) replicates on matched checkpoints but is achieved equally by random bases — it lives on the geometry rungs of the ladder — while our effect-magnitude criterion is null for every basis including SAEs. Registered caveats: our matched Burgers converged (0.010–0.016) where theirs plateaued (0.207); our Allen–Cahn family is milder (ε=0.05 vs. their 10^{-4}). Two findings reconcile the programs independently: their §4.8 replicates our rank-collapse-with-convergence regularity from the failure side (failed PINNs higher-rank, well-trained lower-rank — the same law we measured, discovered from two directions), and our level ladder — association → predictivity → effect-magnitude specificity → spatial localization — assigns the two studies' evidence to different rungs, which is a reconciliation, not a dispute. R2–R5 remain preregistered; R2 (the causal battery on the high-rank Fourier PINN) is the one regime where specificity could still emerge.

7 What To Do Instead

Monitors. Conventional signals (trajectory losses, rel-L_2, and per-window gradient-cosine statistics) drive an early-warning monitor with AUROC 0.875 [0.814, 0.960] on a run-level held-out split under the real

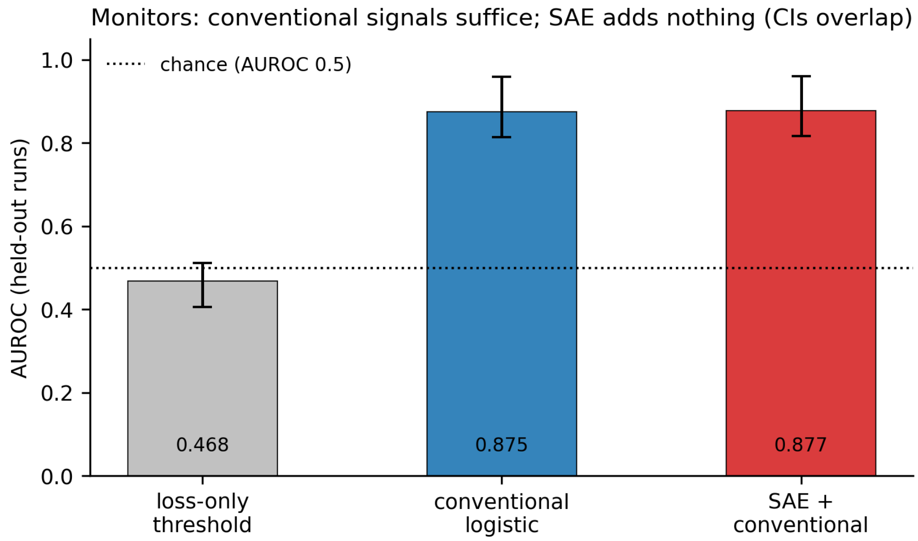

Figure 9: **Early-warning monitors.** Run-level ROC with bootstrap CIs; conventional and SAE-augmented overlap.

feature-window leakage invariant; adding SAE features moves it to 0.877 with P(SAE better) = 0.51 — no established advantage. The loss-only threshold is near chance (0.468) (Figure 9).

Controller. A deterministic state-machine controller (IDLE→WARNING→CONFIRMED→COOLDOWN; the action fires on CONFIRMED entry) executes bounded reweighting actions with rollback through the trainer's per-step lambda seam: on the config-faithful protocol it rescues boundary starvation from 0.421 to 0.0002, matching the oracle reweighting (0.00023). The monitor-source ablation is the punchline: an SAE-feature monitor and a matched *random*-feature monitor drive bit-identical rescue trajectories (0.000129 both) — the SAE features are inert cargo in the closed loop.

SOTA baselines. On the same failure, NTK-adaptive weighting [20] reaches 0.0064; GradNorm [19] (2.01) and RBA [23] (2.19) actively harm. Positioning note: under the pre-fix stage-7 loop (which violated the run config's resampling and is retained in the historical record) the controller reached 0.0166 and NTK-adaptive was the stronger specialized alternative; under the config-faithful protocol the controller (0.0002) beats all three baselines. A single-failure comparison is not optimization-SOTA evidence in either direction; the interpretability claims are kept structurally separate from the engineering claims (Figure 10).

Generalization: the three-failure battery (H17). Because a single-failure rescue is exactly the fragility an earlier review exposed, H17 (preregistered) ran the controller, no-action, oracle, NTK-adaptive, and GradNorm across the three reproducible failure classes — boundary starvation, spectral suppression, and a 2D reaction-diffusion pilot — at three seeds each, behind a machinery gate (the controller must reproduce the v3.4 rescue on the reference seed before any other cell is read; it did — 0.421 → 0.000162). The verdict, exactly as preregistered, is **H17a**: the controller *wins its home class* (boundary starvation 0.000119, beating NTK-adaptive's 0.0044) and *loses to the specialized method elsewhere* (spectral suppression: NTK 0.0036 vs. controller 0.335, whose per-seed spread 0.005/0.995/0.004 shows the bounded λ_{bc} rebalance is unreliable there; RD-2D: NTK 1.67 vs. controller 1.98). It never degrades catastrophically below no-action on any class (worst mean Δ +0.019), and GradNorm actively harms on starvation (3.82 vs. 0.221). One class requires an explicit caveat: on the RD-2D pilot *every* arm, including the oracle reweighting, ends near 1.96 relative error — the task is unlearned at 2500 steps in this regime, so that row measures that no method rescues it, not that any method wins it. The registered reading: the controller's value is never-catastrophic robustness plus home-class dominance — not per-failure SOTA. NTK-adaptive is the best generalist across the tested

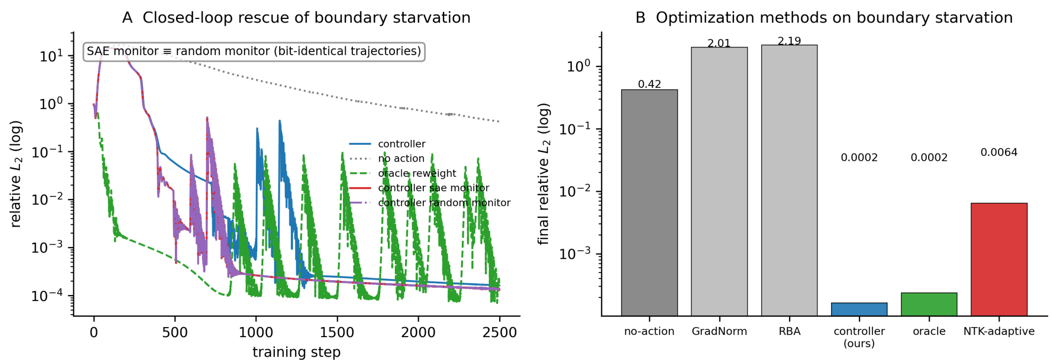

Figure 10: **Controller rescue + SOTA comparison.** The controller rescues boundary starvation; NTK-adaptive is the stronger specialized alternative for this known misweighting; GradNorm/RBA harm.

set.

The monitor floor, corrected (H19). The monitor section's published floor — a single-threshold loss rule at AUROC 0.468 — made the conventional arm look stronger than it is. A preregistered audit (H19) replaced it with an honest baseline: a logistic on past-only loss channels (loss, $\mathcal{L}_{\text{pde}}$, $\mathcal{L}_{\text{bc}}$; no validation-grid rel L_2, no gradients) reaches **0.786** [0.615, 0.898] — closing 78.2% of the threshold→conventional gap. Most of the conventional arm's apparent advantage was the *rule* (threshold vs. trajectory), not the *features* (gradients add 0.786 → 0.875 with overlapping CIs). The same audit traced every monitor-training label to its source: the stage-10 RD-2D runs never entered monitor training (a structural exclusion, not a mislabeling), two pool runs carrying init-transient artifact labels were excluded per the registered rule, and collocation-starvation runs derive no failure step under the operational labeler — the monitor never tested that class. The no-SA-advantage conclusion (P = 0.51) is unchanged.

Monitor	AUROC	95% CI	AUPRC
Loss-only threshold	0.468	[0.406, 0.511]	0.233
Conventional logistic	0.875	[0.814, 0.960]	0.721
SAE + conventional	0.877	[0.817, 0.960]	0.723

Method	Final rel L_2 (boundary starvation)
No action	0.4208
Closed-loop controller (ours)	0.0002
Oracle static reweight	0.0002
GradNorm	2.0134
NTK-adaptive weighting	0.0064
RBA (residual attention)	2.1918

Quantity	Value
Minimum detectable sign-agreement at 80% power ($n = 11$ runs/feature)	85.7%
P(SAE-augmented monitor > conventional)	0.51

The precheck. Before training any SAE on a scientific model's activations: one covariance eigendecomposition of a hidden-layer sample gives $\rho = \text{PR}/W$. If $\rho \ll 0.1$: use linear diagnostics (PCA loadings, supervised probes) for description and gradient/residual signals for prediction and control — and do not expect direction-specific causal effects from *any* basis, sparse or otherwise. If $\rho \gtrsim 0.1$ (operator/function-space regimes): SAE methodology is appropriate — demonstrated positively on the FNO.

8 Discussion and Limitations

- **Scope.** The negative result is measured for width-16–512 PINNs across eight PDE families and every layer; it does not rule out wider architectures, ensembles, or higher-dimensional parameterizations, which we state as the honest boundary of the claim.

- **Corrected measurements.** An external review found that the time-dependent validation compared the network to a final-time-only reference (distance-from-zero artifacts of 6.1/6.0, corrected to 0.111/0.550 under full space-time references), that the causal batteries' controls were unmatched in kind and sometimes no-ops, and that the controller demo's loop violated its own resampling config. All affected stages were re-run; no headline verdict flipped, three numbers moved materially, and both protocol readings are recorded (`docs/external_review_response.md`).

- **The operator causal result is suggestive, not confirmed.** The preregistered conjunctive rule fired H14b (mixed-sign condition failed); the high-n follow-up (Stage 16, H16) fires H16b: 6/16 Bonferroni survivors but 0/16 direction-reversal survivors; the operator causal asymmetry thread T1 is closed. The regime boundary's reconstruction side remains strong (PR 6.8, SAE beats PCA 4.7×); its causal side is closed at H16b.

- **Power.** The causal batteries can detect $\geq$85.7% sign-agreement effects at 80% power; subtler effects would require $\approx$4× the checkpoints. We report this as a design limitation, not a hidden weakness.

- **NTK bridge.** The correlational gradient-conflict $\leftrightarrow$ feature-activity bridge (preregistered H15a/H15b) fired the null: 0/8 features survive Bonferroni and the best association ($|\rho| = 0.578$) exceeds the random-direction 95th percentile (0.494) but fails the preregistered Bonferroni criterion ($p = 0.56$). Gradient conflict is also *chronic* in the starvation regime (75–100% of steps), so the binary conflict label has little within-run variation there; a continuous-magnitude redesign is registered future work. A duplicated-step logging artifact that initially produced degenerate $\rho = \pm 1$ values was caught and fixed before the verdict was read.

- **Controller.** Not optimization SOTA (NTK-adaptive wins on its target failure); its value is failure-agnosticism, bounded actions, and rollback — and H17 generalizes this honestly: home-class dominance (0.00012, beating NTK 0.0044) and never-catastrophic behavior across the battery, but losses to NTK-adaptive on spectral suppression (0.335 vs. 0.0036, with high per-seed variance) and the RD-2D pilot (where every method, including the oracle, fails: the task is unlearned at 2500 steps).

- **Fourier result scope.** H18 is one task (1D Poisson), one embedding family, one width, three seeds. It demonstrates that the regime boundary *can* occur within PINNs, not that every Fourier-feature PINN crosses it; the frequency dose response (R3), a second PDE (R5), and the causal extension (R2) are preregistered follow-ups.

- **The PR bar is operational, not universal.** PR > 3.0 is this protocol's preregistered move-bar, calibrated against the measured width-scaling envelope (1.14–2.51). It is not a universal SAE-suitability cutoff; the continuous ρ-response is exactly what R3 measures.

- **Tangent rank vs. covariance rank.** The Fourier runs dissociate the two (PR 4.0–5.6 with tangent rank 1). We call the participation ratio *effective covariance dimensionality*; it is not the manifold dimension, and interpreting it as such would misstate the geometry.

- **Theory.** The tangent-rank bound is exact; predicting the covariance participation ratio from task and optimizer remains open. The time-dependent PINN PR (1.3–1.9, *below* the steady 2D value) is an intriguing anomaly — the training explored an effectively one-parameter family of states — worth a dedicated study.

- **External validity.** All experiments are 1–10 seeds per configuration on a single-GPU campaign; the preregistration is in-repo rather than third-party (OSF archival is planned).

9 Conclusion

Sparse-feature mechanistic interpretability is not uniformly appropriate across scientific neural networks — and, symmetrically, not uniformly *inappropriate* either. Its usefulness tracks a measurable property of the representation. In conventional low-rank tanh PINNs, sparse and linear feature bases alike show no direction-specific causal information (local tangent rank = input dimension everywhere; covariance PR 1.14–2.51; depth lowers it further). The failure is not intrinsic to PINNs: a Fourier-feature PINN on the same task crosses into the higher-rank regime, and the *same* TopK SAE that was null there before beats matched PCA by 25–56× — a within-PINN boundary, predicted in advance by the rank diagnostic, and dissociating covariance rank from manifold dimension. Concurrent PhysSAE [16] reports positive alignment and spatial concentration in a different layer/dictionary regime; the reconciliation — a level ladder from association to effect-magnitude specificity, a convergent rank-collapse regularity discovered from both sides, and a preregistered head-to-head on matched checkpoints — is itself a contribution. Separately, failure-aware control rescues some PINN optimization failures (home-class dominance, never-catastrophic robustness; not per-failure SOTA), and SAE features are not responsible for that rescue. **Check the geometry before you sparse-code:** one covariance eigendecomposition tells you which regime you are in.

Reproducibility statement. All numbers in this paper are regenerated from committed JSON artifacts (`runs/`) by `scripts/generate_tables.py` and `generate_figures.sh`; a CI job (`scripts/check_results_grounded.py`) fails the build if any results document cites a number ungrounded in an artifact. 171 unit tests; Dockerfile pinned; seeds documented per run; full campaign ≈ 6 GPU-hours (RTX-class). Preregistrations were committed before each stage ran; outcomes are recorded exactly as the decision rules fired.

References

[1] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[2] Sifan Wang, Yujun Teng, and Paris Perdikaris. Understanding and mitigating gradient flow pathologies in physics-informed neural networks. In *SIAM Journal on Scientific Computing*, volume 43, pages A3055–A3081, 2021.

[3] Aditi Krishnapriyan, Amir Gholami, Anima Zewi, Michael W Mahoney, and Karen Willcox. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[4] Nelson Elhage, Tristan Hume, Catherine Olsson, Neel Nanda, et al. Toy models of superposition. In *Anthropic Technical Report*, 2022.

[5] Hoagy Cunningham, Aidan Ewart, Logan Riggs, et al. Sparse autoencoders find highly interpretable features in language models. In *NeurIPS Interpretability Workshop*, 2023.

[6] Trenton Bricken, Adam Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Anthropic Technical Report*, 2023.

[7] Leo Gao, Tom du Tour, Ben Rager, et al. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025.

[8] Samuel Marks, Max Tegmark, et al. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. *arXiv preprint*, 2025.

[9] Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Inducing causal structure for interpretable neural networks. In *International Conference on Machine Learning*, 2022.

[10] Milo Qi and Chris Earls. Mechanistic interpretability of neural quantum states via sparse autoencoders. In *arXiv preprint*, 2026.

[11] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, et al. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[12] DeepMind Mechanistic Interpretability Team. Negative results for sparse autoencoders on downstream tasks. *DeepMind Technical Report*, 2025.

[13] William Leask et al. Sparse autoencoders do not find canonical units of analysis. *arXiv preprint*, 2025.

[14] Gabriel Paulo and Nina Belrose. Sparse autoencoders match or outperform supervised models for concept erasure. *arXiv preprint*, 2025.

[15] Alexander Korznikov et al. Sparse autoencoders discard information relevant to target concepts. *arXiv preprint*, 2026.

[16] Nandita N. Patil, R. A. Eshwar, and Gajanan V. Honnavar. Physsae: Mechanistic interpretability with sparse autoencoders. *arXiv preprint arXiv:2609.07061*, 2026. Concurrent work; penultimate-layer SAEs on six time-dependent PDE families reporting physical-observable alignment and spatially concentrated causal footprints.

[17] Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, Yao Xiao, and ZhengMa Zheng. Frequency principle: Fourier analysis sheds light on deep neural networks. *Communications in Computational Physics*, 28(5):1746–1767, 2020.

[18] Nasim Rahaman, Aristide Barber, Ceyhun Gökmen, Devansh Arpit, Andreas Trischler, and Yoshua Bengio. On the spectral bias of neural networks. In *International Conference on Machine Learning*, 2019.

[19] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*, 2018.

[20] Sifan Wang, Xinling Yu, and Paris Perdikaris. When and why PINNs fail to train: A neural tangent kernel perspective. *Journal of Computational Physics*, 449:110768, 2022.

[21] Levi D. McClenny and Ulisses M. Braga-Neto. Self-adaptive physics-informed neural networks using a soft attention mechanism. *Journal of Computational Physics*, 474:111753, 2023.

[22] Sifan Wang, Hanwen Sankaran, and George Perdikaris. Respecting causality for training physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 421:116813, 2024.

[23] Sokratis Anagnostopoulos et al. Residual-based attention for physics-informed neural networks. *arXiv preprint*, 2024.

[24] Youngkyu Kim, Yoonsang Choi, Dirk Widemann, and Tarek Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, 2022.

[25] Yizhou Chen et al. A toy model of superposition phase transitions. *arXiv preprint*, 2023. PLACEHOLDER — the Bayesian/dynamical phase-transition analysis of the Elhage toy model; verify the actual authors, title, and identifier against the literature before submitting.

[26] Zhengxuan Wu et al. AXBench: Steering language models with sparse autoencoders. In *arXiv preprint*, 2025.

[27] Atticus Geiger, Christopher Potts, Thomas Icard, et al. Neural abstractions. In *Advances in Neural Information Processing Systems*, 2021.

[28] Lu Lu, Pengzhan Jin, Guofei Pang, et al. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[29] Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *NeurIPS*, 2020.